Engineering Mechanics (II)

Dynamics

Bodies at rest .....



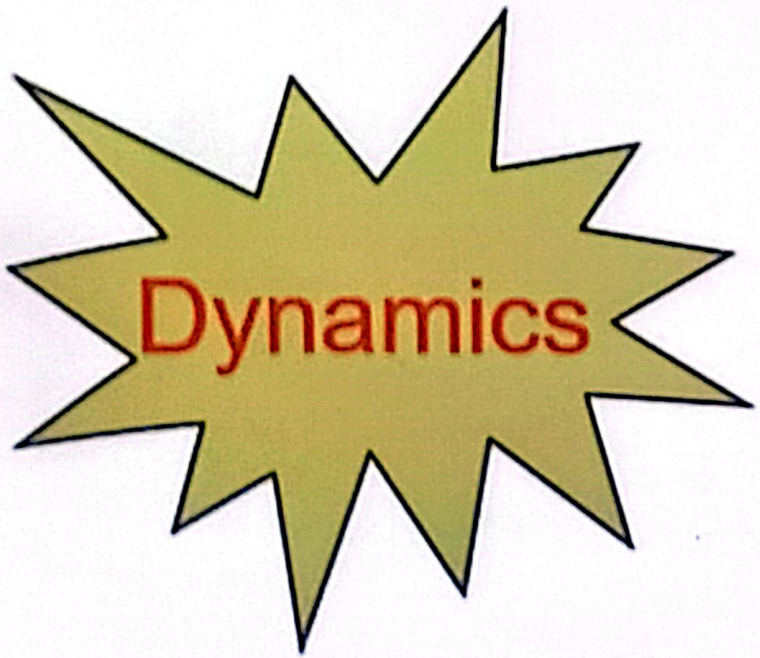
STATICS



Bodies at rest .....

STATICS

Bodies in motion .....



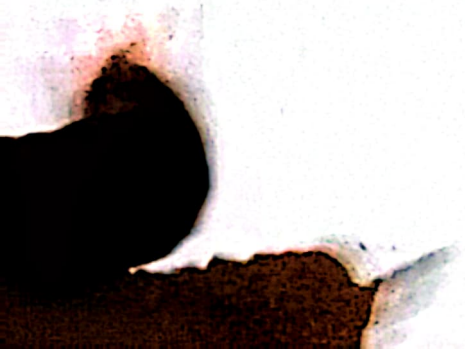
Kinematics

Kinematics:

Study of motion without reference to forces producing motion

Kinematics:

دراسة الحركة بدون الاشارة الي القوى المؤثرة



Kinetics:

Study of the forces acting on a body, and the resulting motion caused by the given forces.



Kinematics

1. Rectilinear Motion

Kinematics

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a. Basic Relations

b. Constant acceleration

Kinematics

1. Rectilinear Motion

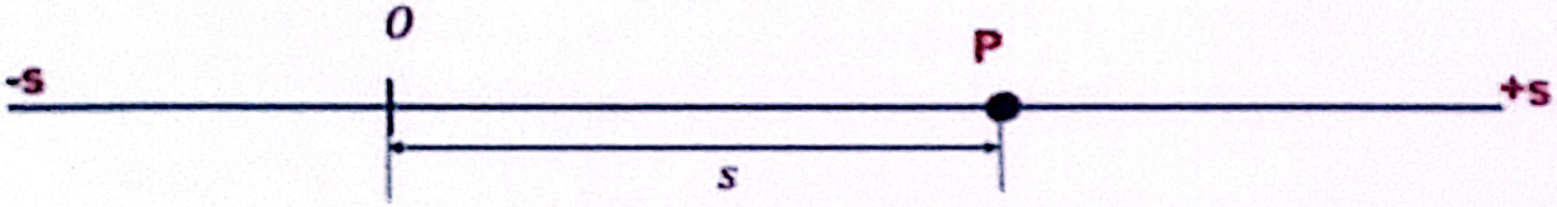
a. Basic Relations

b. Constant acceleration

c. Variable acceleration

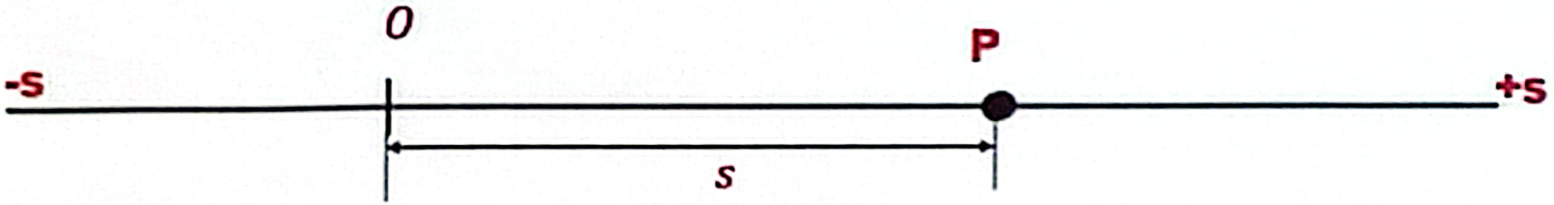
Position:

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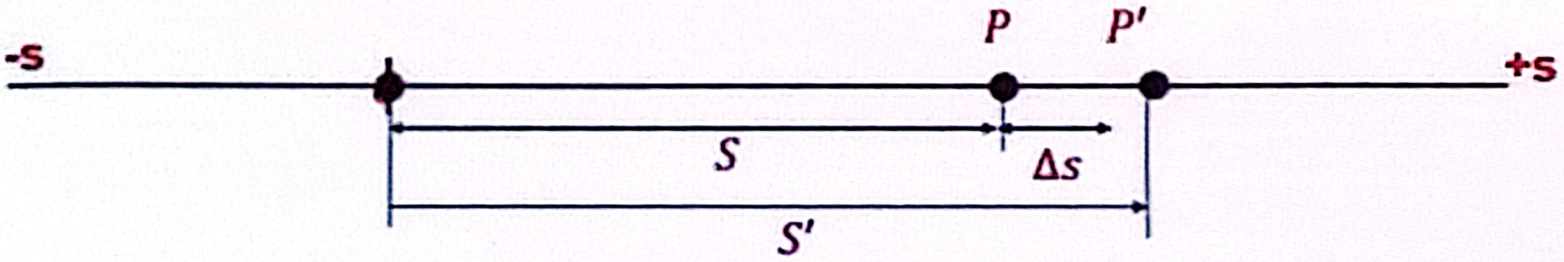


The position S from the origin O at any time t

$$S(t) = f(t)$$

Displacement: is the change in position

If the particle moves from P to P' in time Δt, then



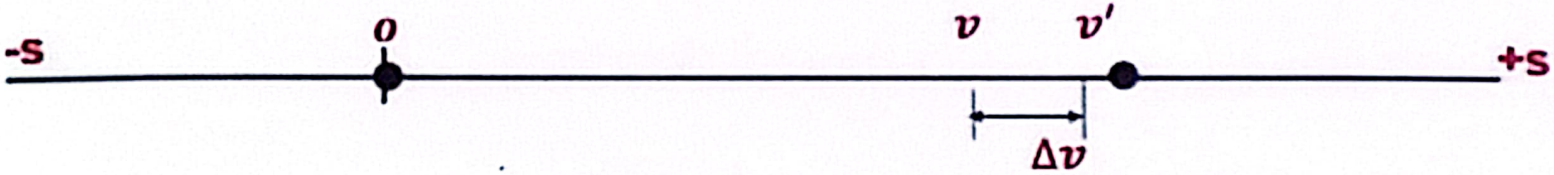
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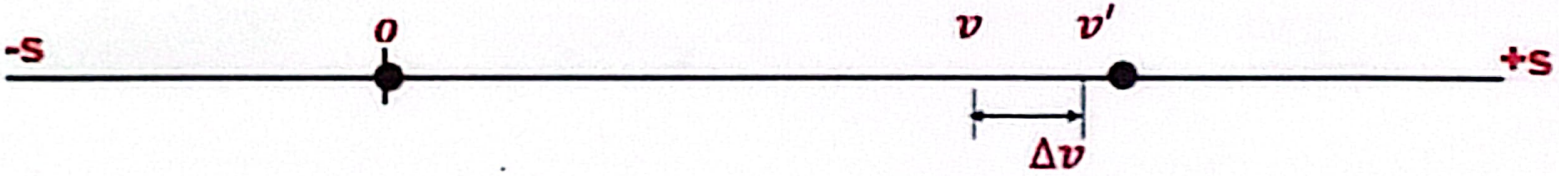


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Rectilinear Motion

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Instantaneous velocity

$$V = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} \longrightarrow v = \frac{dS}{dt}$$

Relation between velocity, position and time:

Instantaneous acceleration

$$a = \lim_{\Delta t \to 0} \frac{\Delta V}{\Delta t} \longrightarrow a = \frac{dv}{dt}$$

Relation between acceleration, velocity and time:

Eliminating time $\longrightarrow a = v\frac{dv}{ds}$

Example:

A particle is moving with a velocity $V = 40 - 5t^2\ m/s^2$

a) The average acceleration from $t = 0$ to $t = 2\ s$:

At $t = 0$: $V_1 = 40\ m/s$

At $t = 2$: $V_2 = 40 - 5(2^2) = 20\ m/s$

$$a_{av} = \frac{\Delta V}{\Delta t} = \frac{V_2 - V_1}{t_2 - t_1} = \frac{20 - 40}{2} = -10\ m/s^2$$

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b) The acceleration at $t = 2\ s$:

c) Position of the particle at any time t:

$$\text{Since } V = \frac{ds}{dt} \longrightarrow dS = V dt$$

$$= (40 - 5t^2)dt$$

$$\therefore \int_0^S dS = \int_0^t V dt = \int_0^t (40 - 5t^2)dt$$

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d) The position of the particle when $t = 2\ s$:

$$\therefore S|_{t=2} = 40(2) - \frac{40}{3} = \frac{200}{3} m/s^2$$

Rectilinear Motion

Position function:

$$x = 6t^2 - t^3$$

